

PIATRA NEAMT, Romania

WMO: 151090

Lat: 46.9336N

Lon: 26.3894E

Elev: 362

StdP: 97.06

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-11.1	-16.1	1.0	-10.5	-14.1	1.2	-9.6	11.8	-2.6	9.0	1.5	1.7	320	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	30.6	19.9	29.1	19.4	27.7	18.9	21.4	28.0	20.6	27.0	19.9	25.9	2.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	14.8	23.9	18.5	14.0	23.0	17.8	13.3	22.2	63.9	28.1	61.1	27.1	58.4	26.1	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	6.8	6.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-2.5	0.7	4.4	9.7	15.7	19.7	20.6	20.9	16.4	10.4	5.1	0.3		
	DBStd	9.20	4.90	5.25	4.92	4.77	3.47	2.88	2.83	3.17	4.32	4.41	4.44	5.51		
	HDD10.0	1410	388	262	181	64	4	0	0	0	3	50	157	302		
	HDD18.3	3258	647	494	433	258	94	19	11	14	85	247	396	559		
	CDD10.0	1467	0	1	6	56	179	291	327	337	195	62	12	1		
	CDD18.3	272	0	0	0	1	12	59	81	92	27	1	0	0		
	CDH23.3	2078	0	0	0	11	104	443	565	728	219	7	0	0		
	CDH26.7	561	0	0	0	1	16	97	158	238	52	0	0	0		
(14) Wind	WSAvg	2.5	2.5	2.5	2.7	3.1	2.7	2.4	2.3	2.4	2.3	2.3	2.2	2.6		
(15) Precipitation	PrecAvg	611	20	21	31	53	74	102	102	62	50	46	28	28		
	PrecMax	935	43	41	63	107	145	256	272	156	119	175	56	59		
	PrecMin	402	3	7	9	10	18	36	24	19	11	3	1	2		
	PrecStd	130	11	9	12	25	38	54	56	32	28	33	15	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	15.3	20.1	24.6	28.5	30.7	32.3	33.1	30.2	24.1	19.0	14.1		
		MCWB	6.8	9.4	10.9	14.6	16.9	21.1	20.5	20.8	18.6	15.4	12.2	8.7		
	2%	DB	8.5	12.4	16.2	22.5	26.1	29.0	30.2	30.9	27.8	21.2	15.8	11.2		
		MCWB	4.7	7.5	9.3	14.1	16.5	19.9	20.1	19.5	17.8	14.4	10.9	6.6		
	5%	DB	6.1	10.0	13.5	20.0	24.3	27.6	28.3	29.3	25.7	19.0	13.6	9.4		
		MCWB	3.1	6.2	7.9	12.5	16.2	19.3	19.5	19.3	17.1	13.2	10.1	5.7		
	10%	DB	4.3	7.2	11.3	17.4	22.3	26.1	26.4	27.5	23.5	16.5	10.9	7.2		
		MCWB	1.9	4.6	6.6	11.6	15.2	18.8	19.1	18.9	16.4	12.3	8.3	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.7	11.8	15.9	19.2	23.6	22.4	22.7	20.2	16.9	13.0	8.8		
		MCDB	11.0	14.4	18.9	23.1	26.3	28.9	29.6	30.0	28.4	21.6	18.3	13.2		
	2%	WB	5.2	7.9	9.9	14.6	18.0	21.2	21.1	21.4	19.0	15.4	11.4	7.2		
		MCDB	7.9	11.7	15.3	21.4	23.8	27.2	27.5	28.3	25.2	19.8	15.0	10.8		
	5%	WB	3.6	6.4	8.4	13.2	17.0	20.3	20.4	20.4	17.8	14.1	10.2	5.9		
		MCDB	5.7	9.1	13.2	19.1	22.1	26.2	26.5	26.9	23.9	18.0	13.1	9.2		
	10%	WB	2.2	4.9	7.1	12.0	16.0	19.3	19.7	19.6	17.0	12.9	8.9	4.6		
		MCDB	4.2	6.9	10.5	16.9	20.8	24.9	25.2	25.7	22.4	15.9	11.0	7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	8.3	9.5	10.0	10.1	10.1	10.6	9.8	8.4	6.3	5.8		
		MCDBR	9.5	10.7	12.6	13.9	12.8	12.4	12.6	13.0	13.2	12.0	10.3	8.9		
	5% WB	MCWBR	6.5	6.6	7.3	7.2	5.6	5.2	5.0	4.8	5.2	6.4	6.0	5.6		
		MCWBR	8.5	9.0	12.1	12.8	11.0	11.5	11.1	11.6	11.9	10.7	9.3	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.345	0.387	0.452	0.425	0.422	0.448	0.457	0.410	0.371	0.340	0.313		
	taud		2.325	2.251	2.201	2.060	2.215	2.261	2.196	2.162	2.255	2.348	2.377	2.354		
	Ebn at Noon		761	807	825	801	842	845	816	788	794	771	724	716		
	Edh at Noon		75	100	122	154	137	132	139	138	115	90	72	66		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.01	3.20	4.39	5.59	5.88	5.91	5.37	3.80	2.45	1.29	0.97		
	RadStd		0.17	0.26	0.34	0.40	0.48	0.52	0.48	0.41	0.42	0.30	0.12	0.11		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (52 neighbors)		+0.65	N/A	N/A	+0.78	N/A	N/A	-145	-202	+80	+18

Nomenclature: See separate page